

System 13: Time-Series Metrics & Monitoring (Prometheus / Datadog Scale)

Zero-Prerequisite Intuition: The "Hospital ICU Heart Monitor" Metaphor What is a Time-Series Database (TSDB), and why can't we just use PostgreSQL? Imagine an Intensive Care Unit (ICU) in a hospital. Every patient has sensors glued to their chest measuring heart rate, oxygen levels, and blood pressure. Every single second, the sensor emits a number: `(09:00:01, 72 bpm)`, `(09:00:02, 73 bpm)`, `(09:00:03, 72 bpm)`.

If you have 500 patients emitting 10 sensor readings every second, you generate **5,000 new rows every second**. In a standard SQL database, every row requires updating B-tree indexes, writing to disk, and managing table locks. In less than 3 days, your database runs out of disk space and crashes!

Furthermore, nobody cares what a patient's exact heart rate was at 03:14:22 AM six months ago. You only care about the **aggregate trend**: *"What was the average heart rate during that hour?"*

A **Time-Series Database (TSDB)** is built specifically for this. It uses specialized **Floating-Point Compression (Gorilla Algorithm)** and **Downsampling Rollups** to compress billions of continuous metrics into tiny chunks of memory.

1. System Scale & Capacity Estimation

Let's design a monitoring infrastructure for an enterprise cloud fleet:

- **Microservices & Hosts:** 10,000 Kubernetes Pods.
- **Metrics per Pod:** 50 distinct metrics (CPU, RAM, HTTP request count, latency percentiles).
- **Total Time-Series Streams:** $10,000 \times 50 = \mathbf{500,000}$ active time series\$.
- **Scrape Interval:** Every 15 seconds.
- **Data Point Ingestion Rate:** $\text{Points / Second} = \frac{500,000}{15} \approx \mathbf{33,333}$ samples/sec\$

2. Distributed Architecture: Pull vs. Push Ingestion

```

flowchart LR
    subgraph Target_Pods ["Kubernetes Microservice Pods"]
        P1["Order Service Pod (Metrics: /metrics)"]
        P2["Payment Service Pod (Metrics: /metrics)"]
        P3["Auth Service Pod (Metrics: /metrics)"]
    end

    subgraph Monitoring_Tier ["Prometheus TSDB Architecture"]
        SD["Service Discovery (Kubernetes API Watcher)"]
        Scraper["Scrape Engine (Pull Worker Pool)"]
        TSDB_Engine["TSDB Head Block (Active RAM Buffer)"]
        WAL["Write-Ahead Log (NVMe SSD)"]
        BlockWriter["2-Hour Block Compactor"]
        ObjectStore["Cold Object Storage (S3 / GCS)"]

        SD -. "Discovers Pod IPs" .-> Scraper
        Scraper -- "HTTP GET /metrics (Every 15s)" --> P1
        Scraper -- "HTTP GET /metrics (Every 15s)" --> P2
        Scraper -- "HTTP GET /metrics (Every 15s)" --> P3

        Scraper --> TSDB_Engine
        TSDB_Engine --> WAL
        TSDB_Engine -. "Every 2 Hours" .-> BlockWriter
        BlockWriter --> ObjectStore
    end
end

```

The Architectural Debate: Pull vs. Push

- **Pull Model (Prometheus):** The central monitoring server initiates an HTTP `GET /metrics` pull from target pods.
 - *Advantage:* The monitoring server controls its own ingestion rate (protects itself from getting overwhelmed); immediately detects when a service is dead if the health scrape times out.
- **Push Model (StatsD / Datadog):** The microservice sends UDP packets outwards to a collection agent.
 - *Advantage:* Ideal for short-lived ephemeral batch jobs (AWS Lambda) that terminate before a pull scraper can reach them.

3. Storage Optimization: The Gorilla Compression Algorithm

Standard 64-bit IEEE-754 floating-point numbers require **8 bytes** each. Storing 33,000 points/sec for a year requires **8.4 Terabytes** of raw numbers. Facebook engineers published the **Gorilla TSDB paper**, demonstrating how to compress 8-byte floats down to an astonishing **1.37 bytes per sample** (83% compression ratio!).

flowchart TD

```
RawFloat["New 64-bit Float Sample: V_curr"] → XOR["XOR with Previous Sample: Δ = V_curr - V_prev"]
XOR → Check{"Is Δ = 0 (Same value)?"}
Check -- Yes → ZeroBit["Store Single '0' Bit (1 bit total!)"]
Check -- No → CheckLeading{"Do Leading & Trailing Zeros match previous?"}
CheckLeading -- Yes → FastBits["Store '10' + Meaningful XOR Bits"]
CheckLeading -- No → VariableBits["Store '11' + New Leading/Length + Meaningful Bit Sequence"]
```

1. Delta-of-Delta Timestamp Compression

Timestamps arrive in predictable 15-second increments ($T_0=0, T_1=15, T_2=30$). Instead of storing the full 64-bit timestamp, compute the **Delta-of-Delta**: $DD = (T_{\text{curr}} - T_{\text{prev}}) - (T_{\text{prev}} - T_{\text{prev2}})$. If the scrape interval is constant (e.g. exactly 15 seconds), $DD = 0$, which is encoded as a **single 0 bit!**

2. XOR Floating-Point Value Compression

Most server metrics change very little from second to second (e.g., CPU is 42.1%, then 42.2%). When two similar floats are XORed, the resulting bit sequence contains a huge string of leading and trailing zeros. Gorilla discards all leading and trailing zeros and writes only the meaningful middle bits.

4. Multi-Tier Retention & Downsampling

Nobody needs 15-second resolution data for an outage that happened 8 months ago. We implement automated downsampling rollups:

```
graph LR
```

```
Raw["Raw Ingestion (15-second resolution)<br>Retention: 7 Days<br>Storage: Local NVMe"]
```

```
FiveMin["5-Minute Rollup (Min, Max, Avg, Count)<br>Retention: 30 Days<br>Storage: Attached SSD"]
```

```
OneHour["1-Hour Rollup (Min, Max, Avg, Count)<br>Retention: 1 Year+<br>Storage: AWS S3"]
```

```
Raw -. "Downsampled after 7d" .-> FiveMin
```

```
FiveMin -. "Downsampled after 30d" .-> OneHour
```

This multi-tier lifecycle allows dashboards to query 1-year historical trends in under 500ms while slashing long-term storage costs by over **95%**.